

MISSING DATA: SOMETHING YOU SHOULD NOT IGNORE

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Missing data

BMJ 2007 ; 334 doi: <https://doi.org/10.1136/bmj.38977.682025.2C> (Published 22 February 2007)

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Almost all studies have some missing observations. Yet textbooks and software commonly assume that data are complete, and the topic of how to handle missing data is not often discussed outside statistics journals.

Guideline

The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) Statement: Guidelines for reporting observational studies ☆

[Erik von Elm](#)^g  , [Douglas G. Altman](#)^b, [Matthias Egger](#)^{a c}, [Stuart J. Pocock](#)^d, [Peter C. Gøtzsche](#)^e, [Jan P. Vandenbroucke](#)^f, [STROBE Initiative](#)

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Original Article

CONSORT 2010 Statement: Updated guidelines for reporting parallel group randomised trials

[Kenneth F. Schulz](#)^a  , [Douglas G. Altman](#)^b, [David Moher](#)^c, [CONSORT Group](#)

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TIM SELL, PT, PhD¹² • RYAN ZEREGA, MS¹² • PAULA DHIMAN, PhD¹³ • RICHARD D. RILEY, PhD¹⁴ • GARY S. COLLINS, PhD¹³

Up Front and Open? Shrouded in Secrecy? Or Somewhere in Between? A Meta-Research Systematic Review of Open Science Practices in Sport Medicine Research

Sports Medicine (2022) 52:2469–2482
<https://doi.org/10.1007/s40279-022-01698-9>

SYSTEMATIC REVIEW



Just How Confident Can We Be in Predicting Sports Injuries? A Systematic Review of the Methodological Conduct and Performance of Existing Musculoskeletal Injury Prediction Models in Sport

Garrett S. Bullock^{1,2} · Joseph Mylott^{1,3} · Tom Hughes^{4,5} · Kristen F. Nicholson¹ · Richard D. Riley⁶ · Gary S. Collins^{7,8}

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Journal of Clinical Epidemiology 165 (2024) 111199

Journal of
Clinical
Epidemiology

ORIGINAL ARTICLE

Open science practices need substantial improvement in prognostic mode studies in oncology using machine learning

Gary S. Collins^{a,1,*}, Rebecca Whittle^{a,1}, Garrett S. Bullock^{b,c}, Patricia Logullo^a, Paula Dhiman^a, Jennifer A. de Beyer^a, Richard D. Riley^d, Michael M. Schluskel^a

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Accepted 20 October 2023; Published online 28 October 2023

Dhiman et al.
Diagnostic and Prognostic Research (2022) 6:13
<https://doi.org/10.1186/s41512-022-00126-w>

Diagnostic and
Prognostic Research

RESEARCH

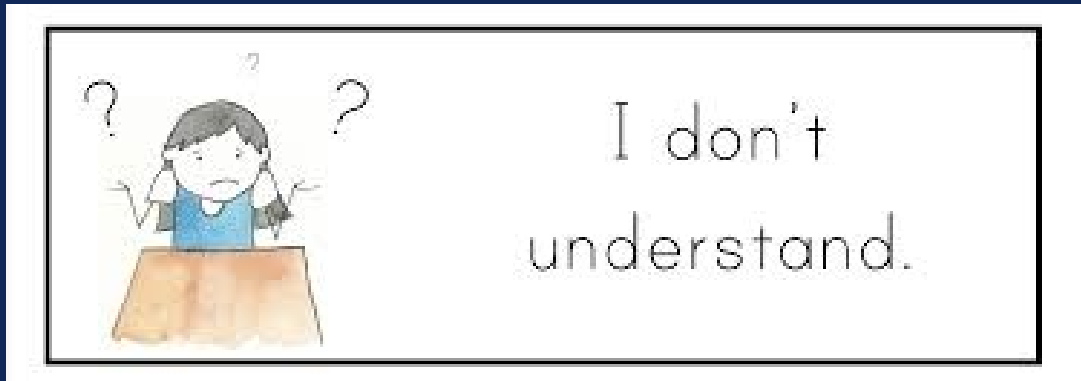
Open Access



Risk of bias of prognostic models developed using machine learning: a systematic review in oncology

Paula Dhiman^{1,2*} · Jie Ma¹, Constanza L. Andaur Navarro^{3,4}, Benjamin Speich^{1,5}, Garrett Bullock⁶, Johanna A. A. Damen^{3,4}, Lotty Hooft^{3,4}, Shona Kirtley¹, Richard D. Riley⁷, Ben Van Calster^{8,9,10}, Karel G. M. Moons^{3,4} and Gary S. Collins^{1,2}





Why Should You Care?

- This is not just an academic exercise
- Has real implications on results

Annoyance in Implications

- Loss of Power
- Loss of Precision in Estimates

Ayilara OF, Zhang L, Sajobi TT, Sawatzky R, Bohm E, Lix LM. Impact of missing data on bias and precision when estimating change in patient-reported outcomes from a clinical registry. Health and quality of life outcomes. 2019 Jun 20;17(1):106.



Patient Implications of Missing Data

- Changes in inferences (switching directions)
- Inability to correctly evaluate cause & effect
- Cannot decipher true relationships

[Home](#) > [Breast Cancer Research and Treatment](#) > [Article](#)

Implications of missing data on reported breast cancer mortality

Epidemiology | Published: 05 November 2022

Volume 197, pages 177–187, (2023) [Cite this article](#)

- Missingness was associated with a greater risk of death:
- NCDB HR 1.23 (99% CI 1.21–1.25)
- SEER HR 2.11 (99% CI 2.05–2.18).

ORIGINAL ARTICLES

Impact of Missing Data Due to Selective Dropouts in Cohort Studies and Clinical Trials

Touloumi, Giota^{1,2}; Pocock, Stuart J.¹; Babiker, Abdel G.³; Darbyshire, Janet H.³

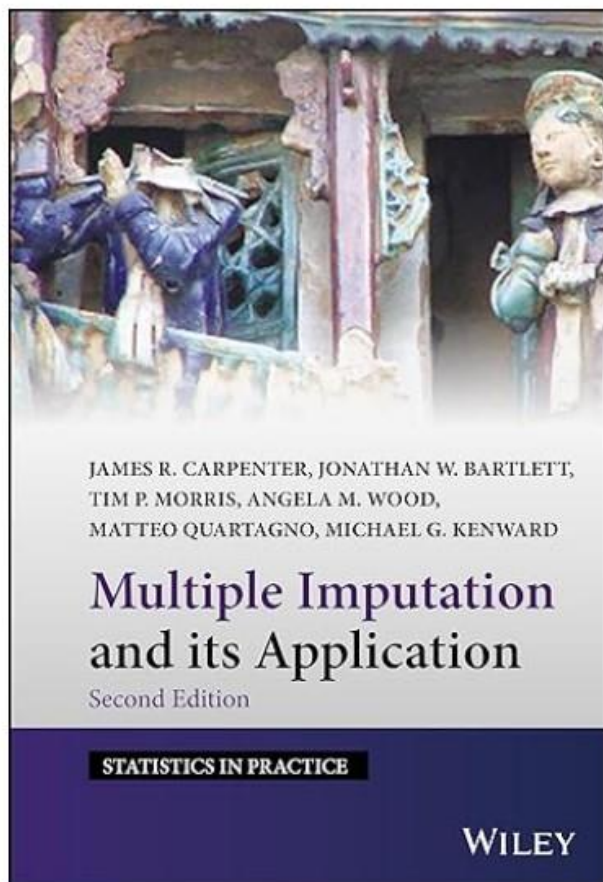
Author Information 

Epidemiology 13(3):p 347-355, May 2002.

- Patients who dropped out were declining worse, thus the trials findings were overly optimistic
- When accounting for drop out, there was no effect!

OK, But What DO We Do About It?

- Evaluate missing prevalence
- Patterns of Missingness
- Mechanisms of Missing
- Decide best route to control for missingness



Read sample

Multiple Imputation and its Application (Statistics in Practice) 2nd Edition

by James R. Carpenter (Author), Jonathan W. Bartlett (Author), Tim P. Morris (Author), & 3 more

3.5 ★★★★★ (2)

Part of: [Statistics in Practice \(49 books\)](#)

[See all formats and editions](#)

Multiple Imputation and its Application

The most up-to-date edition of a bestselling guide to analyzing partially observed data

In this comprehensively revised Second Edition of *Multiple Imputation and its Application*, a team of distinguished statisticians delivers an overview of the issues raised by missing data, the rationale for multiple imputation as a solution, and the practicalities of applying it in a multitude of settings.

With an accessible and carefully structured presentation aimed at quantitative researchers, *Multiple Imputation and its Application* is illustrated with a range of examples and offers key mathematical details. The book includes a wide range of theoretical and computer-based exercises, tested in the classroom, which are especially useful for users of R or Stata. Readers will find:

Step One: Evaluate Prevalence of Missingness

- Overall % missing
- Per variables

STEP ONE: EVALUATE PREVALENCE OF MISSINGNESS

- 'Naniar' Package
- Overall % missing
`sum(is.na(dataframe))`
`dim(dataframe)`
 $(\text{missing\#} / (\text{X\#} * \text{Y\#})) * 100$



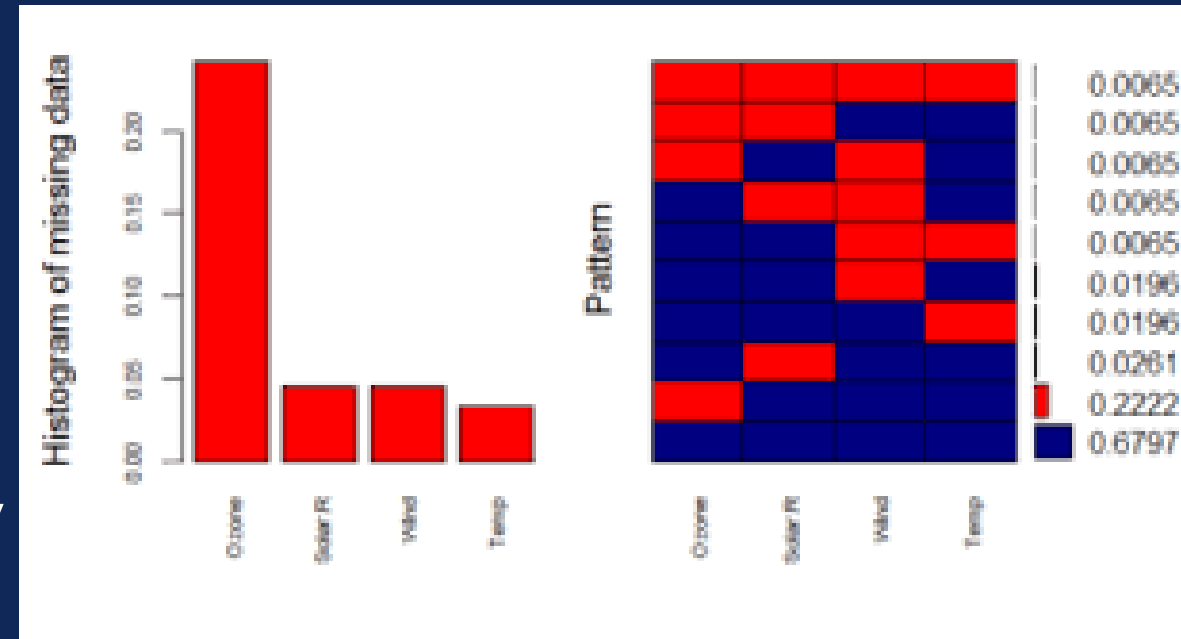
Step One: Evaluate Prevalence of Missingness

Per variables

```
dataframe %>%  
  miss_var_summary() %>%  
  View()
```

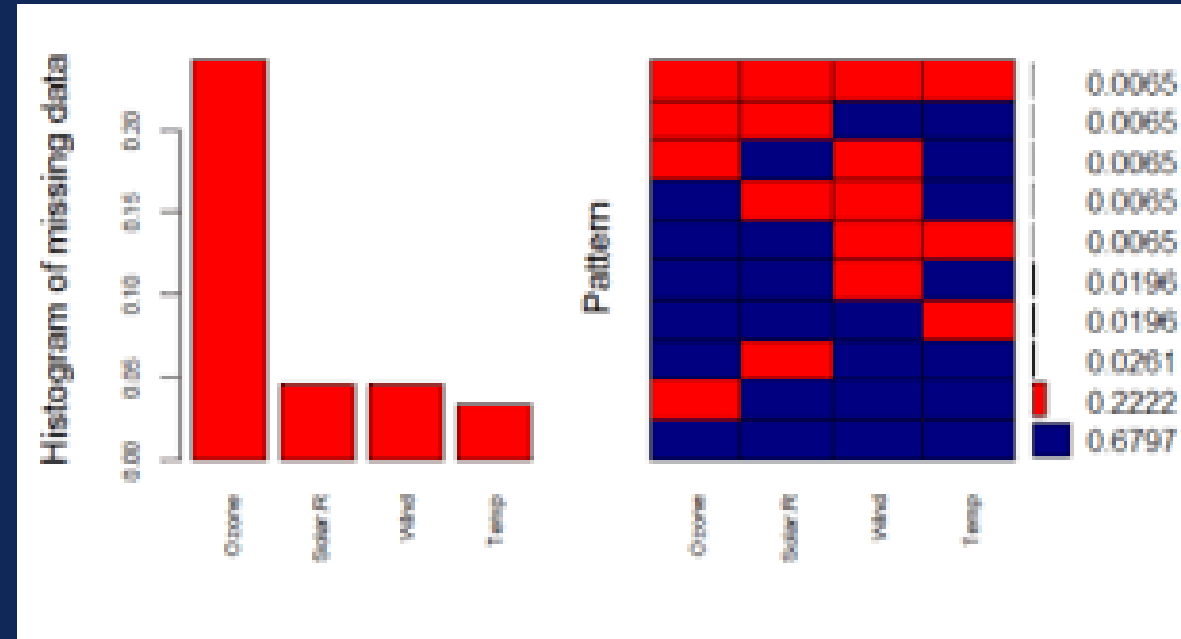
Step Two: Evaluate Patterns of Missingness

- Clustered variables?
- Different time points?
- Changes in missingness by groups?



Step Two: Evaluate Patterns of Missingness

- Clustered variables?
 - `vis_dat(dataframe)`
- Different time points?
 - `ggplot(dataframe,`
 - `aes(x, y)) + geom_miss_point()`
- Changes in missingness by groups?
 - `vis_miss(dataframe)`



Step Three: Determine Mechanism of Missingness

- What can we explain for missingness by
By the data at hand?
- Can we be provided context from the
data collectors?
- Three kinds of missing data mechanisms
 - Missing Completely at Random (MCAR)
 - Missing at Random (MAR)
 - Missing Not at Random (MNAR)

Missing Completely at Random (MCAR)

- There are no patterns... 'random'
- Easiest to control for as it does not induce bias
- Only lack of precision
- VERY RARE, almost never happens

Missing at Random (MAR)

- There are missing data patterns, but it is explained by the data at hand
- Can be controlled for, and should be controlled for
- Induces lack of precision and bias without being controlled for

Missing Not at Random (MNAR)

- There are missing data patterns, but it is NOT explained by the data at hand
- Very difficult to control for,
- Induces lack of precision and bias

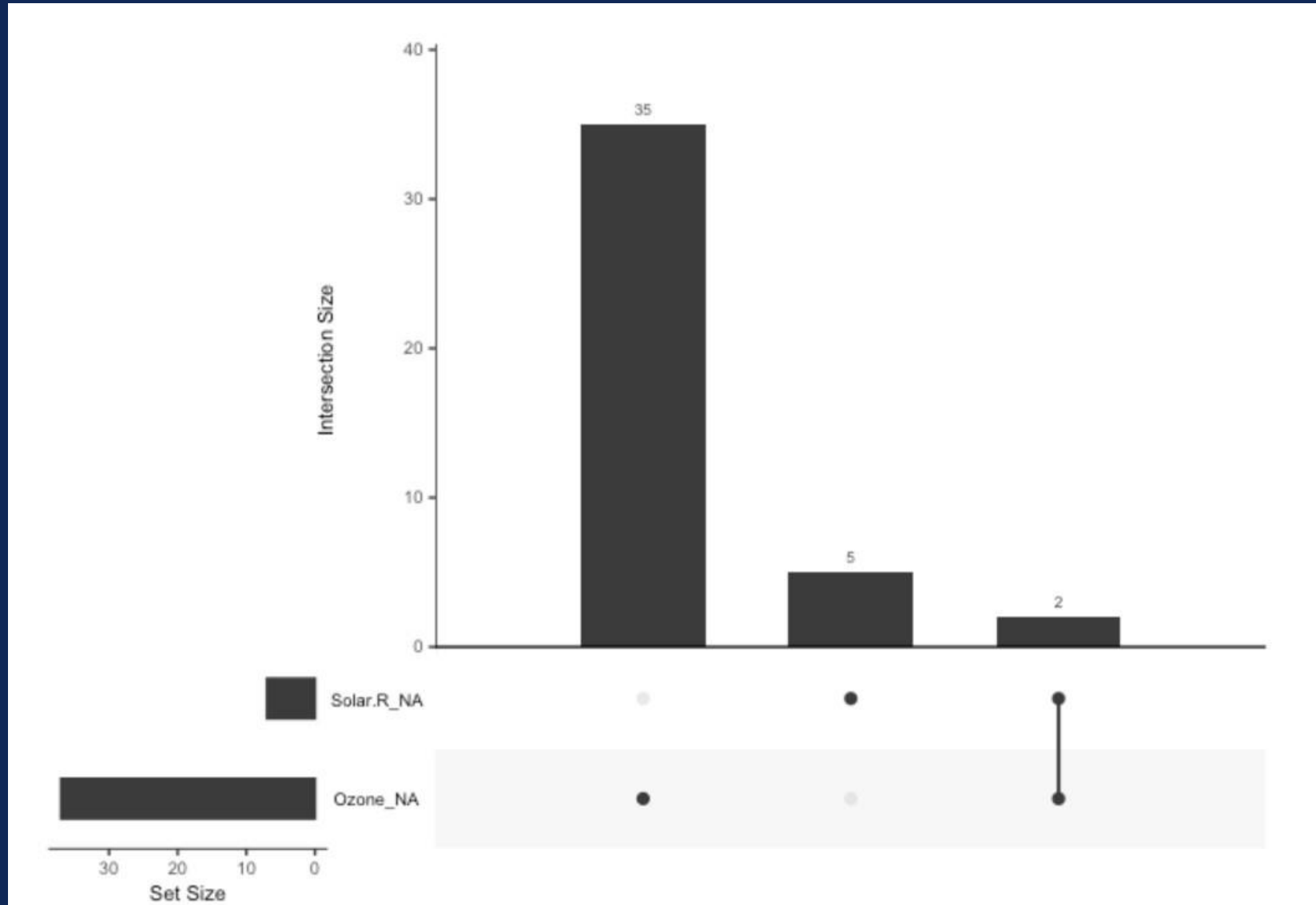
So, What Do We Do About It?

- REPORT your missingness
 - Much of the data will be in the appendices
- <5% total prevalence, perform complete case
- >5%, methods to control for missingness for MAR & MNA

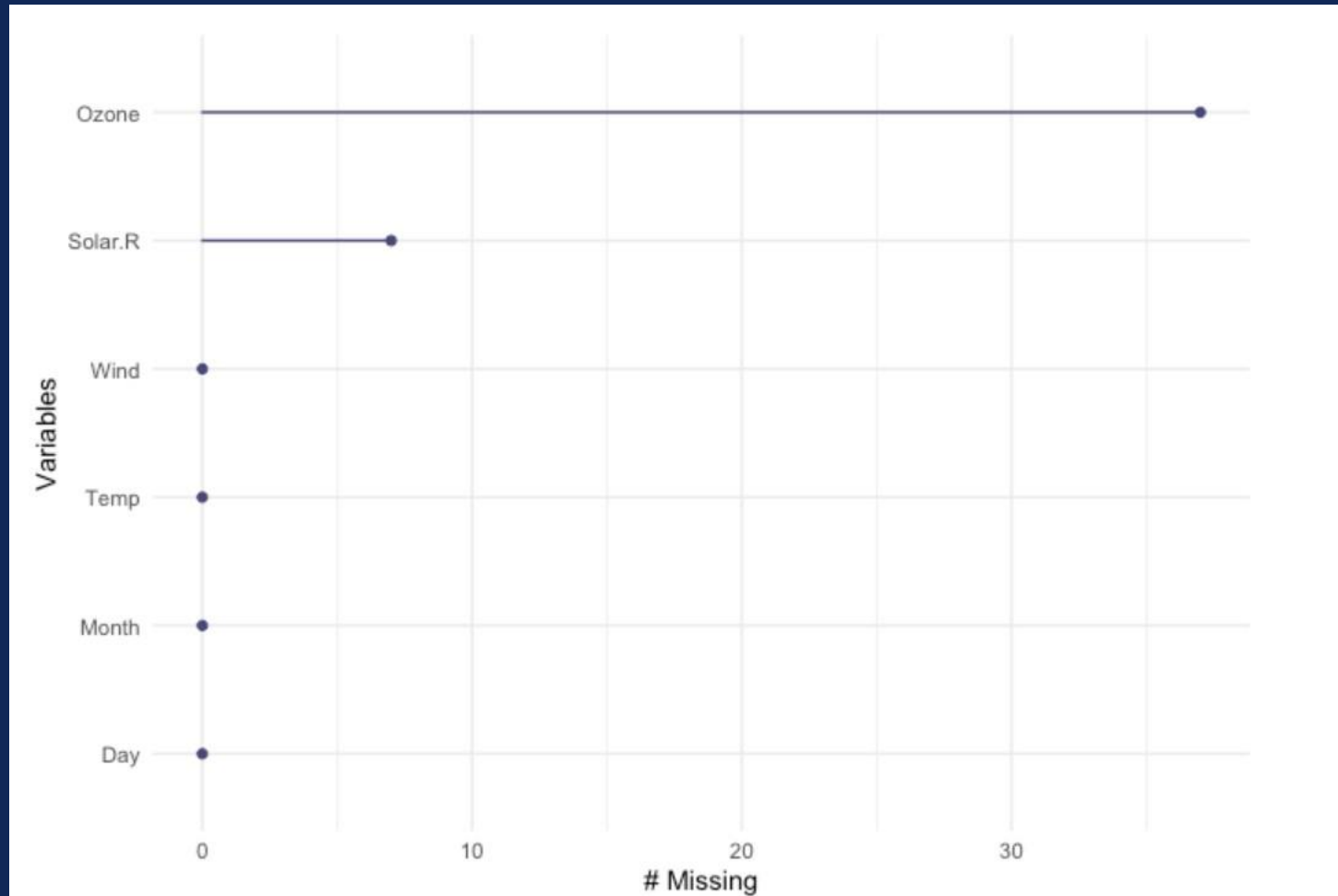
```
miss var summary(airquality)
```

```
#> # A tibble: 6 × 3  
#>   variable n_miss pct_miss  
#>   <chr>     <int>    <num>  
#> 1 Ozone      37     24.2  
#> 2 Solar.R     7      4.58  
#> 3 Wind        0      0  
#> 4 Temp        0      0  
#> 5 Month       0      0  
#> 6 Day         0      0
```

```
gg_miss_upset(airquality)
```



```
gg_miss_var(airquality)
```

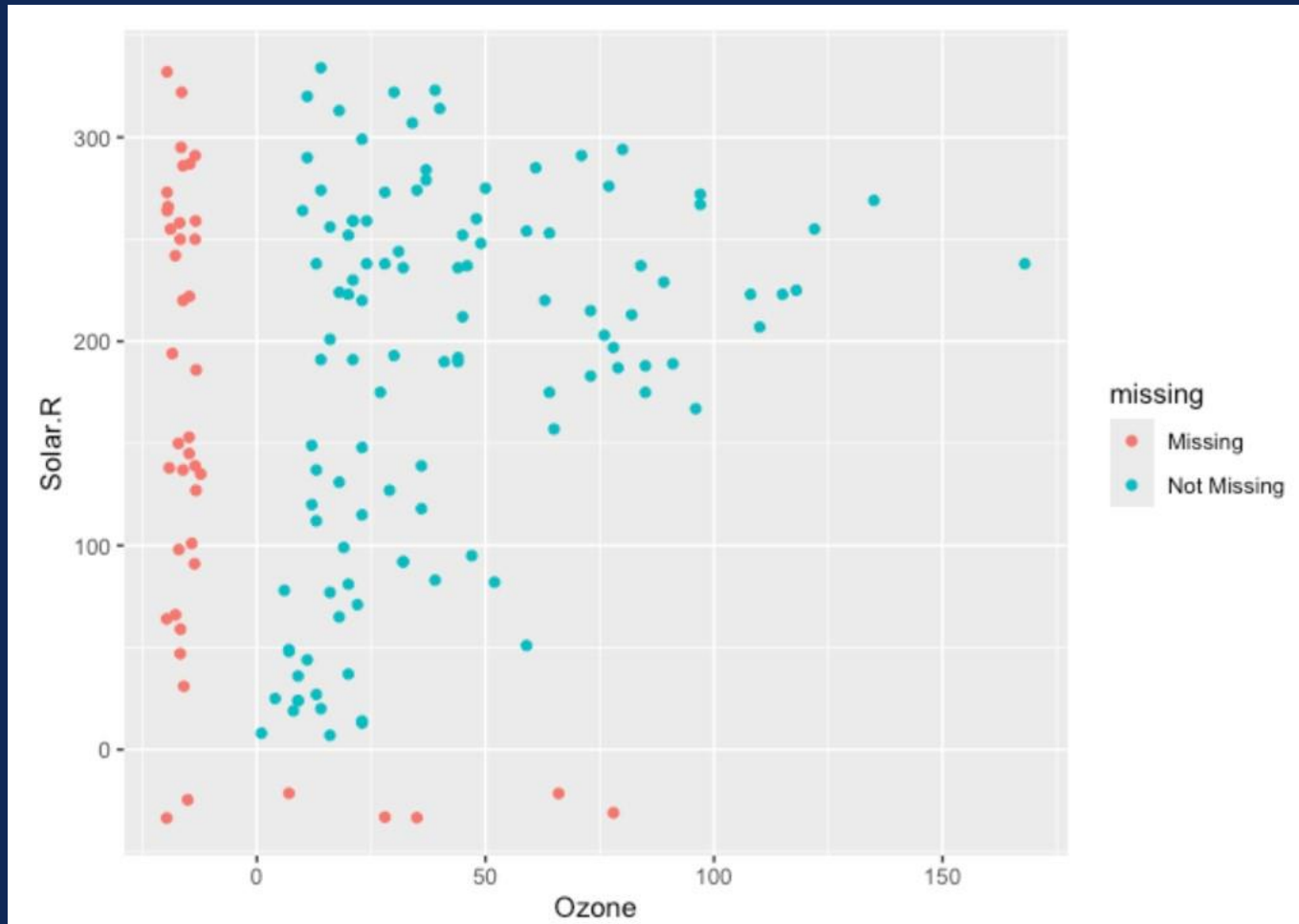


LOOKING AT MISSIGNESS BY MONTH

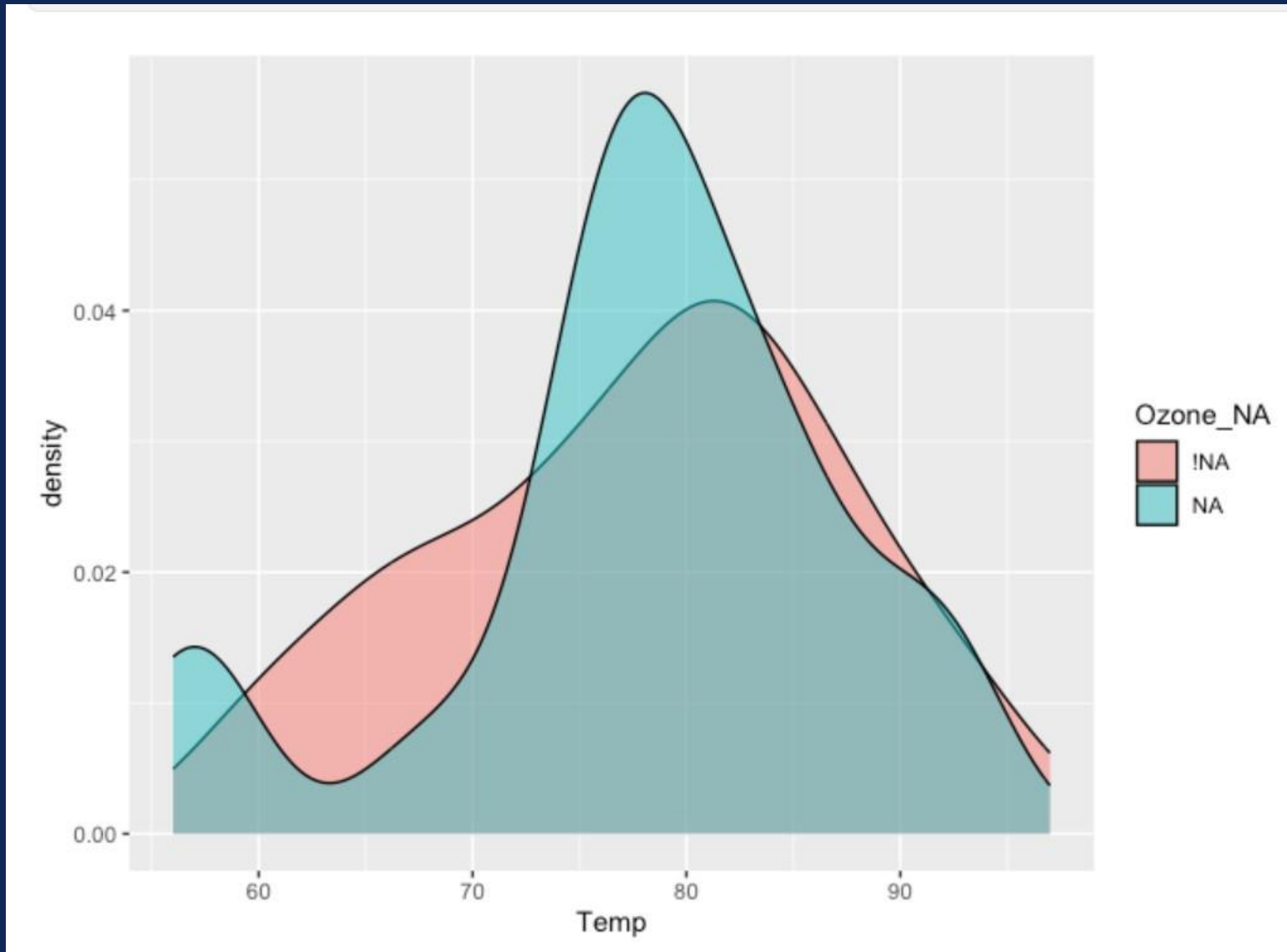
- `airquality %>%`
- `group_by(Month) %>%`
- `miss var summary()`

```
#> # A tibble: 25 x 4
#> # Groups:   Month [5]
#>   Month variable n_miss pct_miss
#>   <int> <chr>      <int>    <num>
#> 1     5 Ozone         5     16.1
#> 2     5 Solar.R       4     12.9
#> 3     5 Wind           0      0
#> 4     5 Temp           0      0
#> 5     5 Day            0      0
#> 6     6 Ozone        21     70
#> 7     6 Solar.R       0      0
#> 8     6 Wind           0      0
#> 9     6 Temp           0      0
#> 10    6 Day            0      0
#> # i 15 more rows
```

```
ggplot(data = airquality, aes(x = Ozone, y = Solar.R)) +  
  geom_miss_point()
```



```
airquality %>% bind shadow() %>%  
ggplot(aes(x = Temp, fill = Ozone_NA)) + geom_density(alpha = 0.5)
```



Beyond the Scope of the Course

- Multiple Imputation
- Bayesian Strategies
- Inverse Probability Weighting
- Propensity Scoring

Final Thoughts

- Missing data is a major problem
- Assess missingness
- Be transparent
- Collaborate with a statistician to help control for missingness